

Practice Questions for Non-basic Variable

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1 Questions

1. In the context of the Simplex Method, what characterizes a non-basic variable?
 - a. It has a positive coefficient in the objective function.
 - b. It is assigned a value of zero in the initial basic feasible solution.
 - c. It is a slack variable introduced to convert inequalities to equalities.
 - d. It has a value greater than zero in the optimal solution.
 - e. It is always associated with a constraint that is binding (active).
2. How does the concept of a non-basic variable relate to the geometric interpretation of the simplex method?
 - a. Non-basic variables define the direction of movement within the feasible region.
 - b. Non-basic variables represent the vertices of the feasible region.
 - c. Setting non-basic variables to zero identifies a vertex of the feasible region.
 - d. Non-basic variables determine the slope of the objective function.
 - e. Non-basic variables are always equal to zero in any feasible solution.
3. During an iteration of the simplex method, why is a non-basic variable selected to enter the basis?
 - a. To ensure feasibility of the solution.
 - b. To maintain non-negativity of all variables.
 - c. To improve the value of the objective function.
 - d. To reach a boundary of the feasible region.
 - e. To simplify the calculations in the simplex tableau.
4. What happens to a non-basic variable when it enters the basis during a simplex iteration?
 - a. Its value remains zero.
 - b. It becomes a basic variable.
 - c. It is eliminated from the problem.
 - d. Its coefficient in the objective function becomes zero.

- e. It is replaced by a slack variable.
5. In the context of the Simplex Method, what is a non-basic variable?
- A variable that is always equal to zero.
 - A variable that is set to zero to obtain an initial feasible solution.
 - A variable whose value is determined by the values of other variables in the system.
 - A variable with a positive coefficient in the objective function.
 - A variable that is never considered for entering the basis.
6. In Slide 7, what is the value of the objective function ζ when we set $x_1 = 0$ and $x_2 = 0$?
- 12
 - 8
 - 10
 - 0
 - 22
7. On Slide 8, when increasing x_1 while keeping $x_2 = 0$, which constraint determines the maximum feasible value of x_1 ?
- $w_1 \geq 0$
 - $w_2 \geq 0$
 - $w_3 \geq 0$
 - $x_1 \geq 0$
 - $x_2 \geq 0$
8. In Slide 11, what is the maximum value of x_2 allowed by the constraints, given that $w_3 = 0$?
- 6.8
 - 8
 - 12
 - 2
 - 5
9. a. After introducing slack variables w_1, w_2, w_3 , what is the equation for w_1 ?
- $w_1 = 12 - x_1 + 3x_2$
 - $w_1 = 12 + x_1 - 3x_2$
 - $w_1 = 12 - x_1 - 3x_2$
 - $w_1 = -12 + x_1 - 3x_2$

- E. $w_1 = -12 - x_1 + 3x_2$
- b. In the initial basic feasible solution (IBFS) obtained by setting the non-basic variables (x_1, x_2) to zero, what is the value of w_2 ?
- A. 0
- B. 3
- C. 5
- D. 8
- E. 12
- c. What is the value of the objective function ζ at the IBFS?
- A. 2
- B. 3
- C. 8
- D. 12
- E. 0
- d. Which variable would enter the basis in the first iteration of the simplex method (using the maximum coefficient rule)?
- A. w_1
- B. w_2
- C. w_3
- D. x_1
- E. x_2
- e. On Slide 8, increasing x_1 while keeping $x_2 = 0$, which constraint determines the maximum feasible value of x_1 ?
- A. $-x_1 + 3x_2 \leq 12$
- B. $x_1 + x_2 \leq 8$
- C. $2x_1 - x_2 \leq 10$
- D. $x_1 \geq 0$
- E. $x_2 \geq 0$
10. In the context of the Simplex Method, what characterizes a non-basic variable?
- a. It has a coefficient of 1 in exactly one equation and 0 in all others.
- b. It is always equal to zero in a basic feasible solution.
- c. It has a positive coefficient in the objective function.
- d. It is a variable introduced to convert inequalities to equalities.
- e. It is a variable that is never changed during the simplex iterations.

2 Answers

1. In the context of the Simplex Method, what characterizes a non-basic variable?
 - a. While a non-basic variable might have a positive coefficient in the objective function, this is not the defining characteristic. The simplex method selects variables to enter the basis based on their coefficients, but the coefficient itself doesn't define whether a variable is basic or non-basic.
 - b. A non-basic variable is set to zero in the initial basic feasible solution. The simplex method iteratively improves the solution by changing which variables are basic and non-basic.
 - c. Slack variables are often non-basic initially, but not all non-basic variables are slack variables. Original decision variables can also be non-basic.
 - d. Non-basic variables can have a value of zero in the optimal solution, but they can also have a value greater than zero in other feasible solutions or in the optimal solution if they become basic variables during the simplex iterations.
 - e. A binding constraint corresponds to a slack variable with a value of zero. Non-basic variables are not directly tied to whether a constraint is binding or not.
2. How does the concept of a non-basic variable relate to the geometric interpretation of the simplex method?
 - a. The direction of movement is determined by which non-basic variable enters the basis, but the non-basic variables themselves don't define the direction.
 - b. Vertices are defined by the intersection of constraints, which are represented by basic variables in the simplex method. Non-basic variables are zero at a vertex.
 - c. In the geometric interpretation, setting all non-basic variables to zero corresponds to selecting a specific vertex of the feasible region. The simplex method moves from one vertex to another by changing which variables are basic and non-basic.
 - d. The slope of the objective function is determined by the coefficients of the variables in the objective function, not specifically the non-basic variables.
 - e. Non-basic variables are zero at a particular vertex, but they can be non-zero at other vertices or feasible solutions.
3. During an iteration of the simplex method, why is a non-basic variable selected to enter the basis?
 - a. Feasibility is maintained by choosing the leaving variable appropriately, not the entering variable.
 - b. Non-negativity is maintained by the choice of the leaving variable, ensuring that all variables remain non-negative after the pivot.
 - c. The primary reason for selecting a non-basic variable to enter the basis is to increase (in maximization problems) or decrease (in minimization problems) the objective function value. The choice is typically based on the variable with the largest positive coefficient in the objective function (for maximization).
 - d. Reaching a boundary is a consequence of the simplex method, not the primary reason for selecting the entering variable.
 - e. While simplifying calculations is a desirable outcome, it's not the primary driver for selecting the entering variable.

4. What happens to a non-basic variable when it enters the basis during a simplex iteration?
 - a. The defining characteristic of a non-basic variable is that its value is zero *before* it enters the basis. Once it enters, its value changes.
 - b. When a non-basic variable enters the basis, it transitions from having a value of zero to becoming a basic variable with a non-zero value (generally). This is the core mechanism of the simplex method's iterative improvement.
 - c. The variable remains part of the problem; its role simply changes from non-basic to basic.
 - d. The coefficient in the objective function might change, but it doesn't necessarily become zero.
 - e. While slack variables are often involved in the simplex method, the entering non-basic variable isn't necessarily replaced by one.
5. In the context of the Simplex Method, what is a non-basic variable?
 - a. While non-basic variables are initially zero, their values can change during the simplex iterations.
 - b. A non-basic variable is initially set to zero to obtain a basic feasible solution. This corresponds to a vertex of the feasible region.
 - c. Basic variables are determined by the values of non-basic variables. Non-basic variables are free to change.
 - d. The coefficient in the objective function influences the choice of entering variable, but doesn't define a non-basic variable.
 - e. Non-basic variables are crucial; the simplex method iteratively selects non-basic variables to enter the basis to improve the objective function.
6. In Slide 7, what is the value of the objective function ζ when we set $x_1 = 0$ and $x_2 = 0$?
 - a. 12 is the constant term in the first constraint equation.
 - b. 8 is the constant term in the second constraint equation.
 - c. 10 is the constant term in the third constraint equation.
 - d. Substituting $x_1 = 0$ and $x_2 = 0$ into $\zeta = 3x_1 + 2x_2$ gives $\zeta = 3(0) + 2(0) = 0$.
 - e. 22 is the optimal value of the objective function, but not at the initial point.
7. On Slide 8, when increasing x_1 while keeping $x_2 = 0$, which constraint determines the maximum feasible value of x_1 ?
 - a. $w_1 = 12 + x_1 - 3x_2 \geq 0$ places no upper bound on x_1 when $x_2 = 0$.
 - b. $w_2 = 8 - x_1 - x_2 \geq 0$ implies $x_1 \leq 8$ when $x_2 = 0$, but this is less restrictive than $x_1 \leq 5$.
 - c. The constraint $w_3 = 10 - 2x_1 + x_2 \geq 0$ becomes $10 - 2x_1 \geq 0$, which implies $x_1 \leq 5$. This is the most restrictive constraint on x_1 in this case.
 - d. This constraint is always satisfied since x_1 is being increased from 0.
 - e. This constraint is explicitly held at 0.
8. In Slide 11, what is the maximum value of x_2 allowed by the constraints, given that $w_3 = 0$?

- a. This is the value obtained from $17 - \frac{5}{2}x_2 = 0$, but $3 - \frac{3}{2}x_2 = 0$ gives a smaller value.
 - b. This is the constant term in the second constraint, unrelated to the current constraints.
 - c. This is the constant term in the first constraint, unrelated to the current constraints.
 - d. From the constraints $w_1 \geq 0$ and $w_2 \geq 0$, we have $17 - \frac{5}{2}x_2 \geq 0$ and $3 - \frac{3}{2}x_2 \geq 0$. The second inequality gives $x_2 \leq 2$, which is the more restrictive constraint.
 - e. This is not directly related to the constraints on x_2 in this iteration.
- 9.
- a. After introducing slack variables w_1, w_2, w_3 , what is the equation for w_1 ?
 - A. This incorrectly uses the wrong signs for x_1 and $3x_2$.
 - B. The correct equation is obtained by adding w_1 to both sides of the first constraint: $w_1 = 12 + x_1 - 3x_2$.
 - C. This incorrectly uses the wrong sign for $3x_2$.
 - D. This incorrectly uses the wrong signs for 12, x_1 , and $3x_2$.
 - E. This incorrectly uses the wrong signs for 12 and x_1 .
 - b. In the initial basic feasible solution (IBFS) obtained by setting the non-basic variables (x_1, x_2) to zero, what is the value of w_2 ?
 - A. This would be true if w_2 was a non-basic variable.
 - B. This is incorrect; it's not related to the given constraints.
 - C. This is incorrect; it's not related to the given constraints.
 - D. With $x_1 = x_2 = 0$, $w_2 = 8 - 0 - 0 = 8$.
 - E. This is the value of w_1 in the IBFS, not w_2 .
 - c. What is the value of the objective function ζ at the IBFS?
 - A. This is incorrect; it's not related to the given objective function.
 - B. This is incorrect; it's not related to the given objective function.
 - C. This is the value of w_2 in the IBFS, not the objective function.
 - D. This is the value of w_1 in the IBFS, not the objective function.
 - E. Since $x_1 = x_2 = 0$ in the IBFS, $\zeta = 3(0) + 2(0) = 0$.
 - d. Which variable would enter the basis in the first iteration of the simplex method (using the maximum coefficient rule)?
 - A. Slack variables are typically basic variables.
 - B. Slack variables are typically basic variables.
 - C. Slack variables are typically basic variables.
 - D. The maximum coefficient rule selects the variable with the largest coefficient in the objective function. In this case, x_1 has a coefficient of 3, which is larger than the coefficient of x_2 (2).
 - E. While x_2 has a positive coefficient, x_1 's coefficient is larger.
 - e. On Slide 8, increasing x_1 while keeping $x_2 = 0$, which constraint determines the maximum feasible value of x_1 ?
 - A. With $x_2 = 0$, this constraint becomes $-x_1 \leq 12$, which is not restrictive.

- B. With $x_2 = 0$, this constraint becomes $x_1 \leq 8$, which is less restrictive than $x_1 \leq 5$.
 - C. Substituting $x_2 = 0$ into $2x_1 - x_2 \leq 10$ gives $2x_1 \leq 10$, so $x_1 \leq 5$. This is the most restrictive constraint on x_1 given $x_2 = 0$.
 - D. This constraint is always satisfied since x_1 is non-negative.
 - E. This constraint is always satisfied since $x_2 = 0$.
10. In the context of the Simplex Method, what characterizes a non-basic variable?
- a. This describes a basic variable in a simplex tableau.
 - b. A non-basic variable is set to zero to obtain a basic feasible solution. The simplex method iteratively changes which variables are basic and non-basic.
 - c. The coefficient in the objective function influences the choice of entering variable, but doesn't define a non-basic variable.
 - d. This describes a slack variable.
 - e. The whole point of the simplex method is to iteratively change which variables are basic and non-basic.